



PROGRAMMABLE WEARABLES FOR HUMAN PERCEPTION

MULTIMODAL ACQUISITION FOR ROAD SAFETY

Guilherme Tavares • João Abrunhosa • Pedro Laredo •
Rafael Soares • Tomás Cruz

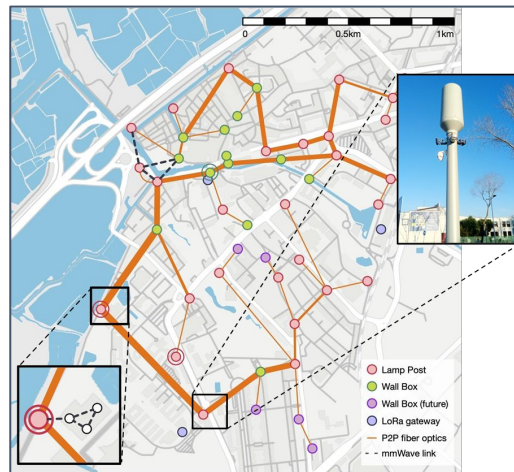
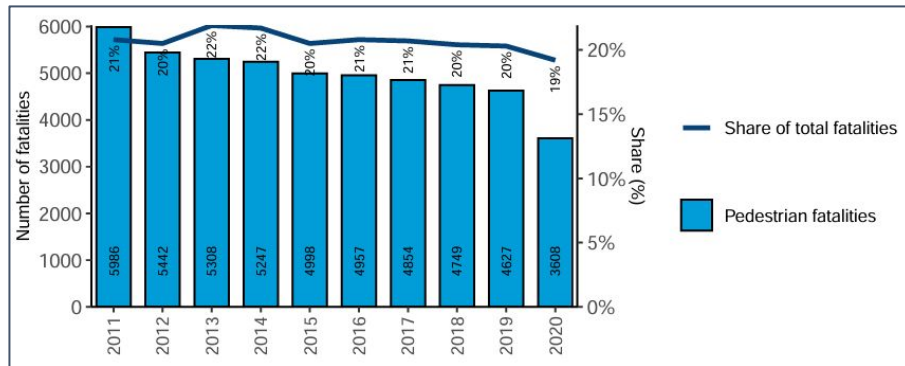
Supervisors: Susana Sargento • Pedro Rito • André
Clérigo • Gonçalo Silva

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Protecting VRUS in Urban Environments

- **VRUs are the main target**
 - Pedestrians remain **highly exposed** in urban traffic scenarios.
- **Infrastructure alone is not enough**
 - **Limited** coverage
 - **Blind spots**
 - High deployment **cost**
- **AR-based solutions validated the concept**
 - **Bulky** AR solutions are not practical for **daily** use
- **Wearables open a more practical path**
 - **Discreet** sensing
 - **Continuous** monitoring
 - **Real-time** alerts



System Architecture – General



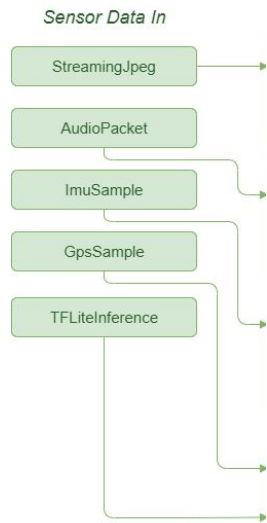
VRU

System Architecture – Wearables



VRU

System Architecture – App



System Architecture – Cloud

From Android Hub

Audio PCM

Camera Frames

Telemetry payload

Glasses pose

Distracted Pedestrian – Propose

- **Microphone** detects a siren + **IMUs** confirm distraction + **GPS/accelerometer** confirm movement → **vibration** on the wrist.



MICROPHONE (GLASSES)

- _SIREN DETECTED
- _MABULANCE APPROACHING



IMU HEAD (GLASSES)

- _NO LATERAL ROTATION DETECTED
- _THE GAZE DINDN'T CHECK THE SIDES



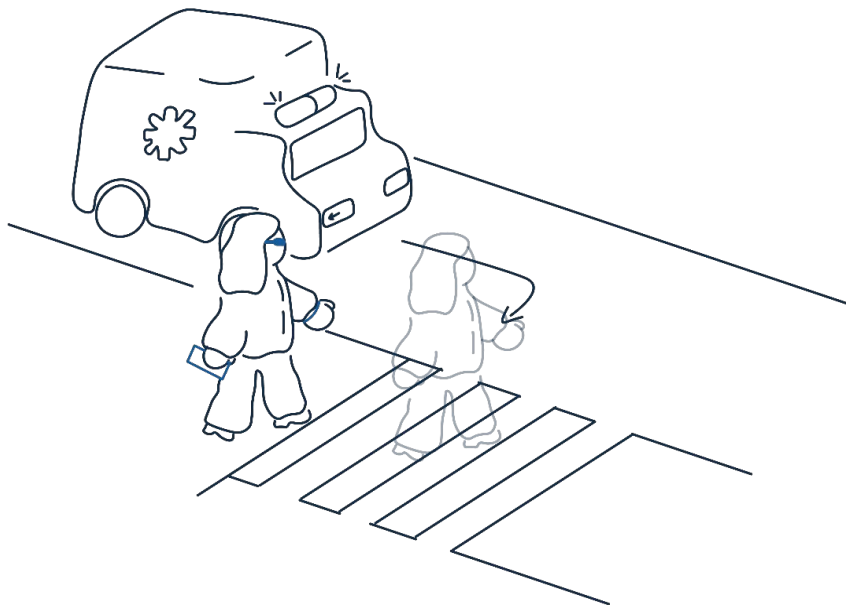
IMU WRIST/ WRISTBAND

- _CONTINUOS MOTION
- _4.8 KM/H
- _NO SLOW DOWN



SMARTPHONE ACCELEROMETER

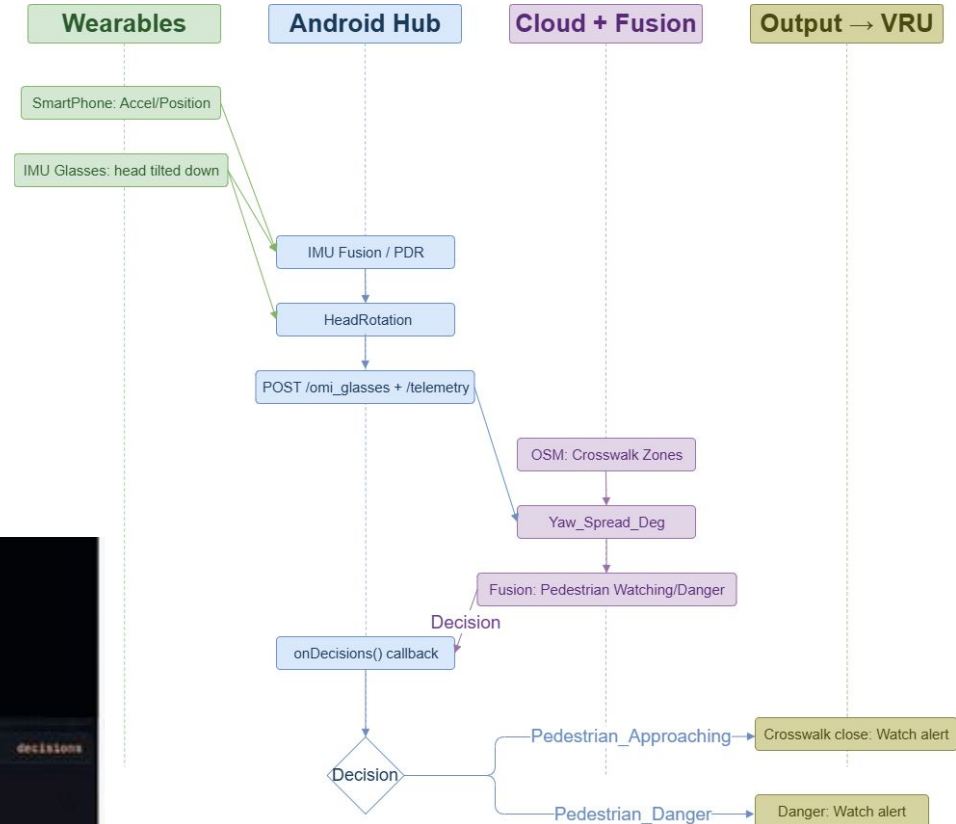
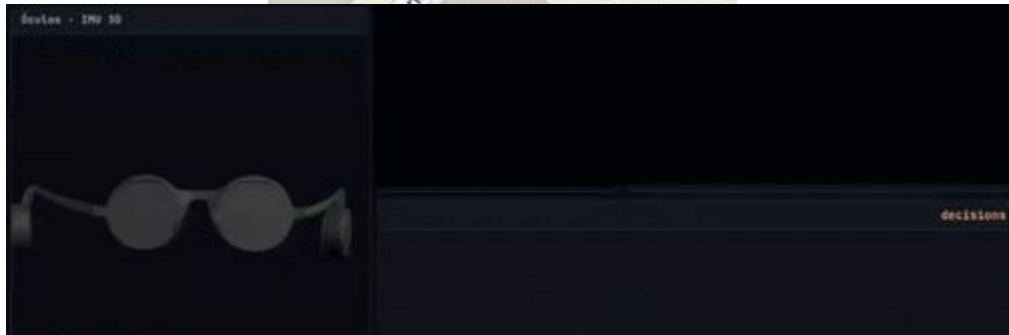
- _ONGOING MOVEMENT
- _WITHOUT STOPPING



TIMELINE UI



Distracted Pedestrian – Result



Crossing Intention – Propose

- Wrist IMU, GPS , and glasses camera infer **crossing intention**, alerting vehicles via **V2X**.



Smart Glasses

**IMU WRIST/ WRISTBAND**

_VELOCITY 0 KM/H
_RELAXED ARM

**DIRECTION OF GAZE**

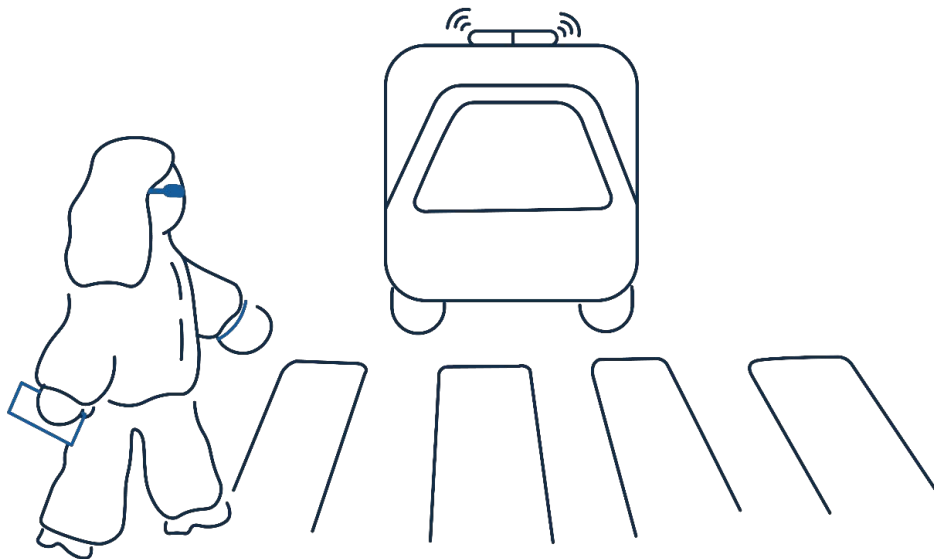
_CARRIAGEWAY

**DISTANCE**

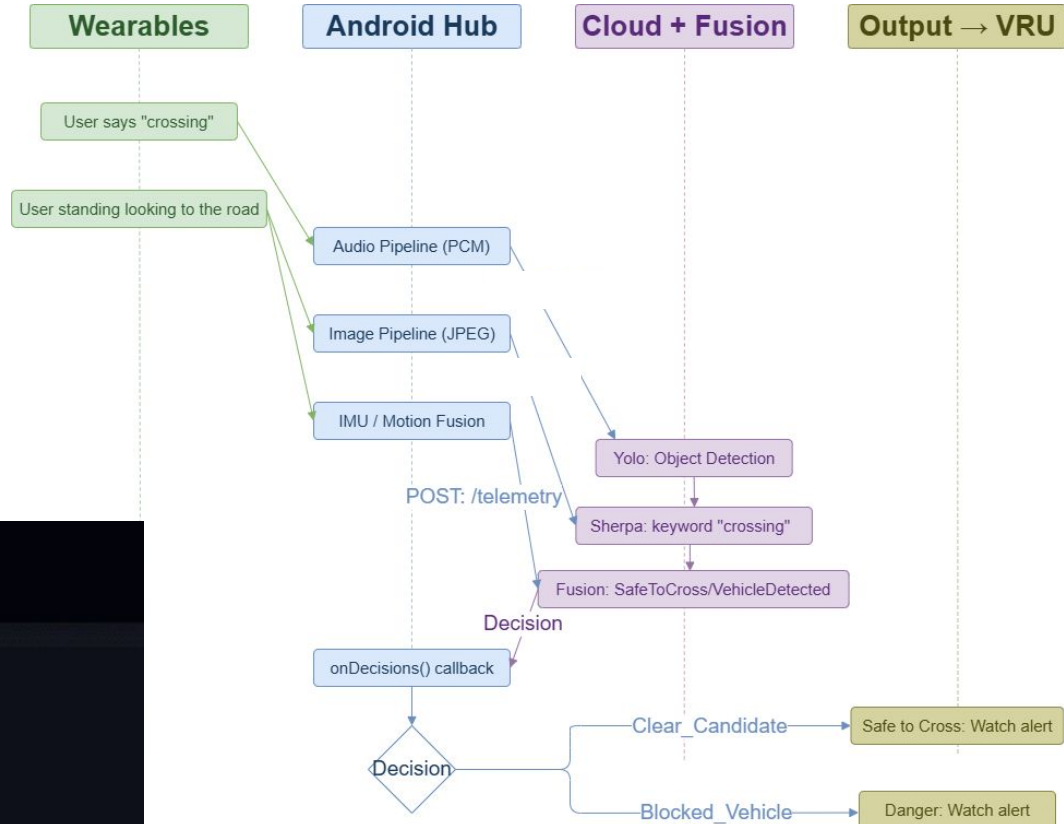
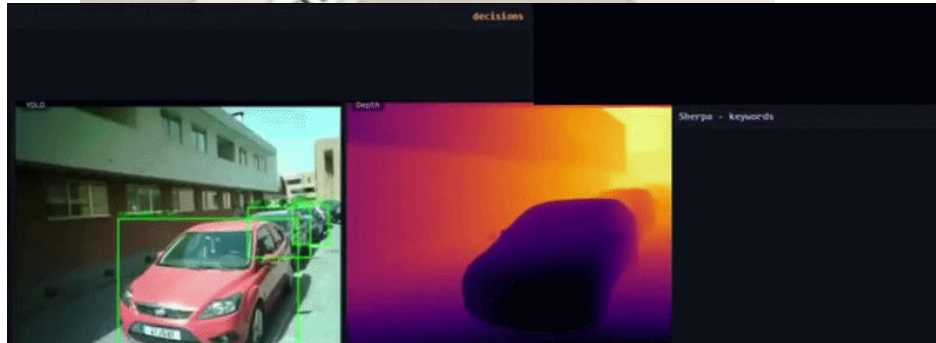
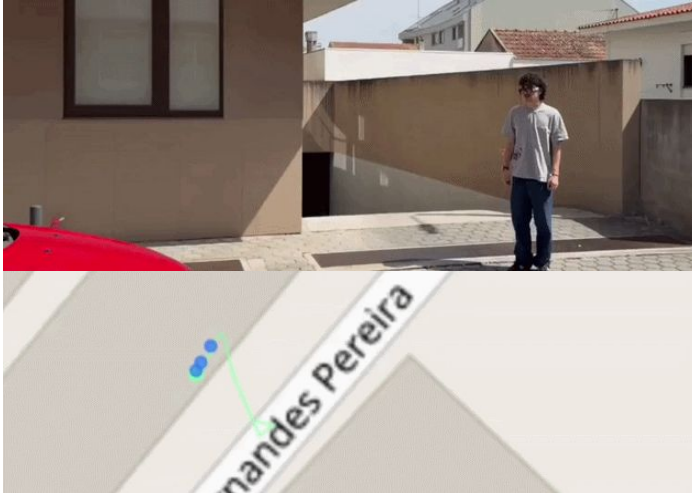
_1.2 M OF CROSSWALK

INTENTION OF CROSS DETECTED

_CONFIDENCE 87%

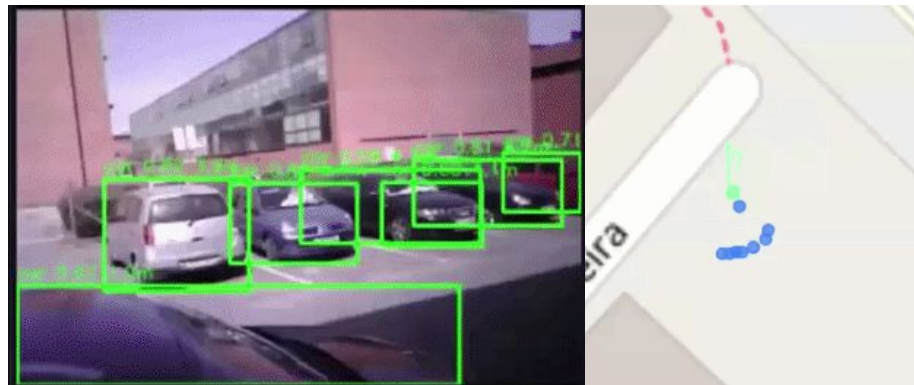


Crossing Intention – Result



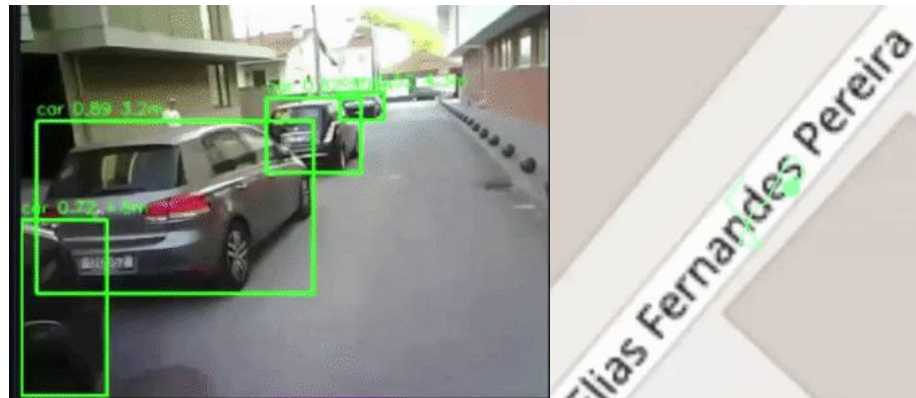
Comparision Ours vs Bsole Firmware

OURS



~20 FPS

BWear



~5 FPS

Thank You for Your Attention

Questions and Answers (Q&A)

